

DECIMAL HEAD TRANSPORT EXPERIMENT

mods: p, N, delta=p-N

samples: solved=82 controls=32 seed=71325

=== $x^3+7 \rightarrow y^2$ ===

-- head2 mappings --

mod p:

solved: total=82 distinct=55

10->10: 8

37->37: 4

11->11: 3

78->78: 3

93->93: 3

96->96: 3

27->27: 2

45->45: 2

48->48: 2

60->60: 2

control: total=32 distinct=24

10->10: 4

80->80: 3

16->16: 2

36->36: 2

61->61: 2

11->11: 1

19->19: 1

21->21: 1

23->23: 1

33->33: 1

mod N:

solved: total=82 distinct=79

10->36: 2

11->21: 2

48->10: 2

10->11: 1

10->35: 1

10->37: 1

10->45: 1

10->55: 1

10->82: 1

10->92: 1

control: total=32 distinct=31

10->11: 2

10->10: 1

10->17: 1

10->61: 1

10->66: 1

11->10: 1

11->68: 1

19->71: 1

21->10: 1

21->46: 1

mod delta:

solved: total=82 distinct=81

40->10: 2

10->15: 1

10->18: 1

11->10: 1

11->20: 1

11->90: 1

13->16: 1

13->25: 1

14->11: 1

14->42: 1

control: total=32 distinct=32

11->29: 1

12->32: 1

12->41: 1

14->15: 1

15->35: 1

15->37: 1

15->38: 1

18->29: 1

20->36: 1

20->41: 1

-- head3 mappings --

mod p:

solved: total=82 distinct=80

```

112->112: 2
930->930: 2
100->100: 1
101->101: 1
103->103: 1
104->104: 1
105->105: 1
106->106: 1
108->108: 1
109->109: 1
control: total=32 distinct=30
108->108: 2
365->365: 2
103->103: 1
106->106: 1
110->110: 1
163->163: 1
166->166: 1
195->195: 1
219->219: 1
234->234: 1

```

```

mod N:
solved: total=82 distinct=82
100->360: 1
100->362: 1
100->370: 1
103->922: 1
105->822: 1
106->456: 1
106->550: 1
107->352: 1
108->114: 1
110->460: 1
control: total=32 distinct=32
100->102: 1
100->111: 1
104->611: 1
104->669: 1
107->115: 1
108->171: 1
112->106: 1
112->683: 1
193->711: 1
216->101: 1

```

```

mod delta:
solved: total=82 distinct=82
100->182: 1
109->154: 1
111->906: 1
114->207: 1
115->107: 1
134->164: 1
138->259: 1
144->112: 1
144->842: 1
146->427: 1
control: total=32 distinct=32
115->293: 1
123->329: 1
128->418: 1
140->150: 1
150->381: 1
155->353: 1
157->378: 1
187->299: 1
205->419: 1
208->363: 1

```

```

=== x^2 -> y^3+7 ===
-- head2 mappings --
mod p:
solved: total=82 distinct=81
10->56: 2
10->10: 1
10->18: 1
10->51: 1
10->52: 1
10->62: 1

```

```

10->63: 1
10->72: 1
10->73: 1
10->81: 1
control: total=32 distinct=32
10->27: 1
10->35: 1
10->46: 1
11->85: 1
13->18: 1
20->62: 1
21->89: 1
28->56: 1
30->10: 1
33->80: 1

mod N:
solved: total=82 distinct=81
10->10: 2
10->13: 1
10->22: 1
10->32: 1
10->66: 1
10->69: 1
10->74: 1
10->95: 1
10->97: 1
11->10: 1
control: total=32 distinct=32
10->21: 1
10->58: 1
11->93: 1
15->11: 1
18->53: 1
23->12: 1
25->10: 1
29->94: 1
33->13: 1
37->12: 1

mod delta:
solved: total=82 distinct=78
12->28: 2
14->13: 2
27->39: 2
28->29: 2
10->14: 1
11->24: 1
11->29: 1
11->70: 1
11->90: 1
13->15: 1
control: total=32 distinct=32
10->39: 1
10->90: 1
15->66: 1
16->42: 1
17->41: 1
18->13: 1
20->18: 1
24->35: 1
25->27: 1
26->39: 1

-- head3 mappings --
mod p:
solved: total=82 distinct=82
100->104: 1
100->567: 1
100->632: 1
100->720: 1
102->522: 1
102->564: 1
102->867: 1
103->815: 1
105->514: 1
105->623: 1
control: total=32 distinct=32
104->271: 1
109->351: 1
109->462: 1

```

```

112->859: 1
137->180: 1
206->625: 1
217->895: 1
285->562: 1
301->106: 1
338->803: 1

```

mod N:

solved: total=82 distinct=82

```

101->742: 1
105->100: 1
106->136: 1
106->952: 1
106->970: 1
107->699: 1
108->105: 1
108->221: 1
108->323: 1
109->664: 1

```

control: total=32 distinct=32

```

105->216: 1
106->589: 1
115->939: 1
151->113: 1
180->534: 1
232->122: 1
257->100: 1
299->942: 1
339->139: 1
377->126: 1

```

mod delta:

solved: total=82 distinct=82

```

109->143: 1
110->908: 1
112->703: 1
113->298: 1
114->240: 1
123->281: 1
126->289: 1
130->226: 1
133->156: 1
135->930: 1

```

control: total=32 distinct=32

```

107->390: 1
107->908: 1
153->664: 1
169->420: 1
171->418: 1
184->139: 1
209->183: 1
249->356: 1
256->275: 1
267->390: 1

```

=== solved lane slices ===

-- x^3+7 -> y^2 --

head2 by lane3:

mod p:

lane3=0: total=27 distinct=24

```

10->10: 4
11->11: 1
14->14: 1
15->15: 1
23->23: 1

```

lane3=1: total=28 distinct=24

```

10->10: 2
37->37: 2
45->45: 2
78->78: 2
12->12: 1

```

lane3=2: total=27 distinct=21

```

10->10: 2
11->11: 2
27->27: 2
37->37: 2
48->48: 2

```

mod N:

lane3=0: total=27 distinct=27

```

10->11: 1
10->36: 1
10->37: 1
11->10: 1
11->44: 1
lane3=1: total=28 distinct=26
11->21: 2
48->10: 2
10->45: 1
10->92: 1
11->22: 1
lane3=2: total=27 distinct=27
10->35: 1
10->36: 1
10->55: 1
10->82: 1
11->48: 1
mod delta:
lane3=0: total=27 distinct=27
14->84: 1
16->27: 1
18->25: 1
18->41: 1
19->44: 1
lane3=1: total=28 distinct=28
10->15: 1
10->18: 1
11->10: 1
11->20: 1
11->90: 1
lane3=2: total=27 distinct=27
13->16: 1
15->11: 1
15->39: 1
17->10: 1
19->87: 1
head2 by lane5:
mod p:
lane5=0: total=26 distinct=22
37->37: 3
10->10: 2
78->78: 2
11->11: 1
13->13: 1
lane5=1: total=14 distinct=14
12->12: 1
15->15: 1
27->27: 1
29->29: 1
32->32: 1
lane5=2: total=14 distinct=13
10->10: 2
11->11: 1
23->23: 1
27->27: 1
44->44: 1
lane5=3: total=14 distinct=12
10->10: 3
11->11: 1
14->14: 1
37->37: 1
48->48: 1
lane5=4: total=14 distinct=14
10->10: 1
20->20: 1
25->25: 1
26->26: 1
30->30: 1
mod N:
lane5=0: total=26 distinct=26
10->45: 1
11->10: 1
11->46: 1
14->30: 1
21->57: 1
lane5=1: total=14 distinct=14
10->36: 1
11->21: 1
11->44: 1
14->11: 1

```

```

18->57: 1
lane5=2: total=14 distinct=14
10->37: 1
10->55: 1
11->21: 1
11->22: 1
16->90: 1
lane5=3: total=14 distinct=14
10->92: 1
11->48: 1
11->95: 1
29->61: 1
33->99: 1
lane5=4: total=14 distinct=14
10->11: 1
10->35: 1
10->36: 1
10->82: 1
12->10: 1
mod delta:
lane5=0: total=26 distinct=26
11->20: 1
11->90: 1
13->16: 1
18->25: 1
18->41: 1
lane5=1: total=14 distinct=14
15->11: 1
20->23: 1
20->94: 1
22->17: 1
22->89: 1
lane5=2: total=14 distinct=14
10->15: 1
11->10: 1
14->11: 1
14->42: 1
14->84: 1
lane5=3: total=14 distinct=14
13->25: 1
15->39: 1
16->27: 1
19->44: 1
21->31: 1
lane5=4: total=14 distinct=14
10->18: 1
17->10: 1
18->16: 1
26->84: 1
27->29: 1

head3 by lane3:
mod p:
lane3=0: total=27 distinct=27
101->101: 1
103->103: 1
108->108: 1
109->109: 1
112->112: 1
lane3=1: total=28 distinct=28
105->105: 1
106->106: 1
127->127: 1
201->201: 1
301->301: 1
lane3=2: total=27 distinct=27
100->100: 1
104->104: 1
112->112: 1
118->118: 1
139->139: 1
mod N:
lane3=0: total=27 distinct=27
100->362: 1
100->370: 1
108->114: 1
114->104: 1
115->447: 1
lane3=1: total=28 distinct=28
103->922: 1
106->456: 1

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110->460: 1
111->219: 1
114->212: 1
lane3=2: total=27 distinct=27
100->360: 1
105->822: 1
106->550: 1
107->352: 1
113->954: 1
mod delta:
lane3=0: total=27 distinct=27
144->842: 1
169->275: 1
181->258: 1
185->419: 1
194->443: 1
lane3=1: total=28 distinct=28
100->182: 1
109->154: 1
111->906: 1
114->207: 1
115->107: 1
lane3=2: total=27 distinct=27
134->164: 1
156->112: 1
158->395: 1
173->107: 1
196->874: 1

head3 by lane5:
mod p:
lane5=0: total=26 distinct=26
100->100: 1
103->103: 1
118->118: 1
139->139: 1
169->169: 1
lane5=1: total=14 distinct=14
127->127: 1
155->155: 1
278->278: 1
298->298: 1
327->327: 1
lane5=2: total=14 distinct=14
104->104: 1
106->106: 1
112->112: 1
232->232: 1
272->272: 1
lane5=3: total=14 distinct=14
101->101: 1
105->105: 1
108->108: 1
112->112: 1
143->143: 1
lane5=4: total=14 distinct=14
109->109: 1
201->201: 1
256->256: 1
263->263: 1
301->301: 1
mod N:
lane5=0: total=26 distinct=26
106->456: 1
110->460: 1
114->104: 1
142->305: 1
210->575: 1
lane5=1: total=14 distinct=14
100->360: 1
111->219: 1
115->447: 1
144->115: 1
183->576: 1
lane5=2: total=14 distinct=14
100->370: 1
106->550: 1
114->212: 1
114->229: 1
161->907: 1
lane5=3: total=14 distinct=14

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103->922: 1
113->954: 1
118->481: 1
294->610: 1
339->999: 1
lane5=4: total=14 distinct=14
100->362: 1
105->822: 1
107->352: 1
108->114: 1
121->101: 1
mod delta:
lane5=0: total=26 distinct=26
111->906: 1
114->207: 1
134->164: 1
181->258: 1
185->419: 1
lane5=1: total=14 distinct=14
156->112: 1
200->947: 1
201->236: 1
221->895: 1
226->174: 1
lane5=2: total=14 distinct=14
109->154: 1
115->107: 1
144->112: 1
144->842: 1
146->427: 1
lane5=3: total=14 distinct=14
138->259: 1
158->395: 1
169->275: 1
194->443: 1
216->772: 1
lane5=4: total=14 distinct=14
100->182: 1
173->107: 1
187->163: 1
266->844: 1
277->299: 1

-- x^2 -> y^3+7 --
head2 by lane3:
mod p:
lane3=0: total=27 distinct=27
10->18: 1
10->51: 1
10->63: 1
10->72: 1
10->81: 1
lane3=1: total=28 distinct=28
10->52: 1
10->56: 1
10->73: 1
18->44: 1
19->87: 1
lane3=2: total=27 distinct=27
10->10: 1
10->56: 1
10->62: 1
10->86: 1
12->20: 1
mod N:
lane3=0: total=27 distinct=27
10->95: 1
11->10: 1
11->17: 1
12->42: 1
20->21: 1
lane3=1: total=28 distinct=28
10->13: 1
10->32: 1
13->94: 1
14->72: 1
20->10: 1
lane3=2: total=27 distinct=26
10->10: 2
10->22: 1

```



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10->66: 1
10->69: 1
10->74: 1
mod delta:
lane3=0: total=27 distinct=26
27->39: 2
10->14: 1
11->29: 1
12->28: 1
14->13: 1
lane3=1: total=28 distinct=28
11->90: 1
12->28: 1
13->15: 1
13->22: 1
13->93: 1
lane3=2: total=27 distinct=27
11->24: 1
11->70: 1
14->86: 1
15->10: 1
18->21: 1

head2 by lane5:
mod p:
lane5=0: total=26 distinct=26
10->18: 1
10->51: 1
10->52: 1
10->56: 1
10->72: 1
lane5=1: total=14 distinct=14
10->56: 1
10->63: 1
10->86: 1
12->10: 1
18->44: 1
lane5=2: total=14 distinct=14
10->10: 1
22->57: 1
34->24: 1
34->77: 1
40->62: 1
lane5=3: total=14 distinct=14
19->88: 1
19->96: 1
22->11: 1
34->89: 1
36->65: 1
lane5=4: total=14 distinct=14
10->62: 1
10->81: 1
12->20: 1
31->37: 1
36->10: 1
mod N:
lane5=0: total=26 distinct=26
10->10: 1
10->69: 1
10->74: 1
10->97: 1
11->52: 1
lane5=1: total=14 distinct=14
10->95: 1
13->94: 1
14->72: 1
19->92: 1
24->72: 1
lane5=2: total=14 distinct=14
10->13: 1
10->22: 1
20->11: 1
20->21: 1
33->26: 1
lane5=3: total=14 distinct=14
10->10: 1
10->32: 1
10->66: 1
11->10: 1
11->17: 1
lane5=4: total=14 distinct=14

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```

22->52: 1
23->14: 1
25->64: 1
26->24: 1
28->64: 1
mod delta:
lane5=0: total=26 distinct=25
14->13: 2
11->24: 1
11->29: 1
12->28: 1
14->16: 1
lane5=1: total=14 distinct=14
11->70: 1
12->28: 1
13->15: 1
13->93: 1
19->38: 1
lane5=2: total=14 distinct=14
10->14: 1
19->34: 1
25->32: 1
28->21: 1
31->24: 1
lane5=3: total=14 distinct=14
13->22: 1
15->10: 1
18->22: 1
23->26: 1
24->14: 1
lane5=4: total=14 distinct=14
11->90: 1
26->19: 1
28->32: 1
28->36: 1
33->51: 1

head3 by lane3:
mod p:
lane3=0: total=27 distinct=27
100->632: 1
100->720: 1
103->815: 1
105->514: 1
107->182: 1
lane3=1: total=28 distinct=28
102->522: 1
102->564: 1
107->739: 1
182->441: 1
198->877: 1
lane3=2: total=27 distinct=27
100->104: 1
100->567: 1
102->867: 1
105->623: 1
129->206: 1
mod N:
lane3=0: total=27 distinct=27
106->952: 1
113->100: 1
114->175: 1
126->423: 1
202->216: 1
lane3=1: total=28 distinct=28
106->136: 1
108->323: 1
131->946: 1
140->720: 1
202->111: 1
lane3=2: total=27 distinct=27
101->742: 1
105->100: 1
106->970: 1
107->699: 1
108->105: 1
mod delta:
lane3=0: total=27 distinct=27
109->143: 1
113->298: 1
123->281: 1

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145->139: 1
243->146: 1
lane3=1: total=28 distinct=28
110->908: 1
126->289: 1
130->226: 1
133->156: 1
135->930: 1
lane3=2: total=27 distinct=27
112->703: 1
114->240: 1
148->866: 1
157->109: 1
186->221: 1

head3 by lane5:
mod p:
lane5=0: total=26 distinct=26
100->720: 1
102->522: 1
102->564: 1
105->514: 1
107->182: 1
lane5=1: total=14 distinct=14
100->567: 1
100->632: 1
102->867: 1
125->104: 1
182->441: 1
lane5=2: total=14 distinct=14
100->104: 1
220->576: 1
341->777: 1
342->240: 1
400->626: 1
lane5=3: total=14 distinct=14
194->968: 1
198->887: 1
228->111: 1
342->891: 1
361->650: 1
lane5=4: total=14 distinct=14
103->815: 1
105->623: 1
129->206: 1
319->370: 1
360->106: 1
mod N:
lane5=0: total=26 distinct=26
101->742: 1
105->100: 1
106->970: 1
107->699: 1
116->524: 1
lane5=1: total=14 distinct=14
106->952: 1
131->946: 1
140->720: 1
192->928: 1
244->725: 1
lane5=2: total=14 distinct=14
106->136: 1
108->221: 1
202->111: 1
202->216: 1
333->262: 1
lane5=3: total=14 distinct=14
108->105: 1
108->323: 1
109->664: 1
113->100: 1
114->175: 1
lane5=4: total=14 distinct=14
227->520: 1
236->143: 1
256->640: 1
260->244: 1
282->770: 1
mod delta:
lane5=0: total=26 distinct=26
113->298: 1

```

```

114->240: 1
123->281: 1
140->139: 1
145->139: 1
lane5=1: total=14 distinct=14
112->703: 1
126->289: 1
133->156: 1
135->930: 1
198->382: 1
lane5=2: total=14 distinct=14
109->143: 1
192->349: 1
257->326: 1
282->217: 1
316->245: 1
lane5=3: total=14 distinct=14
130->226: 1
157->109: 1
186->221: 1
234->260: 1
243->146: 1
lane5=4: total=14 distinct=14
110->908: 1
268->196: 1
287->368: 1
288->328: 1
339->515: 1

=== sample solved rows ===
P65 l3=2 l5=0 mod p      eq1 h2=13->13  h3=139->139  eq2 h2=18->10  h3=187->101
P65 l3=2 l5=0 mod N      eq1 h2=62->80  h3=621->800  eq2 h2=10->69  h3=107->699
P65 l3=2 l5=0 mod delta eq1 h2=20->26  h3=201->268  eq2 h2=37->17  h3=377->170
P66 l3=0 l5=1 mod p      eq1 h2=53->53  h3=532->532  eq2 h2=10->63  h3=100->632
P66 l3=0 l5=1 mod N      eq1 h2=11->44  h3=115->447  eq2 h2=10->95  h3=106->952
P66 l3=0 l5=1 mod delta eq1 h2=39->28  h3=398->289  eq2 h2=34->11  h3=345->112
P67 l3=1 l5=2 mod p      eq1 h2=49->49  h3=499->499  eq2 h2=69->73  h3=693->733
P67 l3=1 l5=2 mod N      eq1 h2=11->22  h3=114->229  eq2 h2=20->11  h3=202->111
P67 l3=1 l5=2 mod delta eq1 h2=25->20  h3=252->201  eq2 h2=86->33  h3=863->339
P68 l3=2 l5=3 mod p      eq1 h2=98->98  h3=985->985  eq2 h2=37->15  h3=371->157
P68 l3=2 l5=3 mod N      eq1 h2=11->95  h3=113->954  eq2 h2=10->66  h3=109->664
P68 l3=2 l5=3 mod delta eq1 h2=28->40  h3=289->406  eq2 h2=15->10  h3=157->109
P69 l3=0 l5=4 mod p      eq1 h2=10->10  h3=109->109  eq2 h2=76->35  h3=769->359
P69 l3=0 l5=4 mod N      eq1 h2=36->53  h3=366->534  eq2 h2=53->53  h3=532->536
P69 l3=0 l5=4 mod delta eq1 h2=35->33  h3=350->334  eq2 h2=58->21  h3=580->217
P70 l3=1 l5=0 mod p      eq1 h2=41->41  h3=411->411  eq2 h2=87->47  h3=877->475
P70 l3=1 l5=0 mod N      eq1 h2=73->89  h3=731->890  eq2 h2=80->50  h3=809->507
P70 l3=1 l5=0 mod delta eq1 h2=39->36  h3=395->363  eq2 h2=31->33  h3=318->332
P75 l3=0 l5=0 mod p      eq1 h2=73->73  h3=730->730  eq2 h2=10->72  h3=100->720
P75 l3=0 l5=0 mod N      eq1 h2=34->37  h3=343->378  eq2 h2=90->24  h3=901->242
P75 l3=0 l5=0 mod delta eq1 h2=40->10  h3=407->107  eq2 h2=14->13  h3=145->139
P80 l3=2 l5=0 mod p      eq1 h2=16->16  h3=169->169  eq2 h2=88->79  h3=888->791
P80 l3=2 l5=0 mod N      eq1 h2=81->60  h3=817->609  eq2 h2=82->42  h3=821->423
P80 l3=2 l5=0 mod delta eq1 h2=32->37  h3=323->377  eq2 h2=18->21  h3=189->218

```

```
=== P135 rx transport worked example ===
```

```
P135 true rx tail ...368
```

```
bridge ry tail ...644
```

```
mod p (curve law):
```

```
x^3 mod p tail ...9834
```

```
y^2 mod p tail ...9841
```

```
y^2 - x^3 = 7
```

```
x^3 + 4 = y^2 - 3 ? True
```

```
x^3 + 7 = y^2 ? True
```

```
p cube-root branches (same x^3 mod p): 3
```

```
mod N (fingerprint transport):
```

```
(x^3+4) mod N tail ...768 (lhs; Wolfram ...768)
```

```
(y^2-3) mod N tail ...962 (rhs; bridge ry)
```

```
lhs == rhs mod N ? False
```

```
law: px^3 + 4 == py^2 - 3 (equiv px^3 + 7 == py^2)
```

```
head2 lhs->24 rhs->39
```

```
head3 lhs->248 rhs->397
```

```
meet cube roots mod N: 3
```

```
meet sqrt roots mod N: 2
```

```
modulus cube lift:
```

```
p^3 mod N == delta^3 mod N ? True
```

```

heaven lift (bridge rx,ry):
    carry ok: True
    y_comp mod N tail ...952
    y_comp == meet ? False

=== next step ===
1. Keep P135 true rx branch (90653...368), not bridge slot 26000...739.
2. N transport:  $rx^3 \bmod N (...764) + 4 \rightarrow \text{meet } ...768$ .
3. p transport:  $x^3 + 7 = y^2$  via bridge ry (...9841).
4. Meet pairs ( $x^3=y^2 \bmod N$ ) are NOT on-curve mod p  $\rightarrow$  use heaven lift, not raw meet coords.
5. Hunt P135 d with shelf2+offset AND filter ( $px^3+4)=(py^2-3) \bmod p$ .
6. On mod N, lhs (...768) and rhs (...962) split – that gap is the N displacement.

```

```

meet sqrt y0 tail ...612 on-curve-p with rx: False
meet sqrt y1 tail ...725 on-curve-p with rx: False

```

```

=== unsolved batch anchor gaps (rx + bridge ry) ===
P135 gap_h2=14 gap_h3=148 lhs...768 rhs...962
P140 gap_h2=11 gap_h3=110 lhs...943 rhs...962
P145 gap_h2=62 gap_h3=629 lhs...794 rhs...962
P150 gap_h2=10 gap_h3=105 lhs...264 rhs...962
P155 gap_h2=69 gap_h3=690 lhs...055 rhs...962
P160 gap_h2=73 gap_h3=733 lhs...898 rhs...962

```

```
rows -> C:\Users\mitch\Desktop\secp256k1\ARCHIVE\decimal_head_transport_rows.tsv
```

DETAILED ROWS

```

group label puzzle_n lane3 lane5 mod eq1_h2 eq1_h3 eq2_h2 eq2_h3
solved P1 1 1 1 p 32->32 327->327 60->28 603->285
solved P1 1 1 1 N 11->21 111->219 63->78 638->784
solved P1 1 1 1 delta 30->56 304->562 12->28 126->289
solved P2 2 2 2 p 10->10 104->104 34->77 341->777
solved P2 2 2 2 N 10->55 106->550 49->11 496->118
solved P2 2 2 2 delta 24->13 243->139 31->24 316->245
solved P3 3 0 3 p 96->96 964->964 74->40 741->405
solved P3 3 0 3 N 46->61 466->613 11->10 113->100
solved P3 3 0 3 delta 35->23 359->231 32->38 321->381
solved P4 4 1 4 p 82->82 829->829 58->71 586->719
solved P4 4 1 4 N 12->10 121->101 40->76 407->761
solved P4 4 1 4 delta 10->18 100->182 26->19 268->196
solved P5 5 2 0 p 71->71 713->713 51->84 512->842
solved P5 5 2 0 N 84->40 840->400 10->10 105->100
solved P5 5 2 0 delta 43->31 430->317 33->41 335->413
solved P6 6 0 1 p 39->39 391->391 69->20 694->200
solved P6 6 0 1 N 86->46 866->466 26->60 260->600
solved P6 6 0 1 delta 28->29 287->294 33->17 336->173
solved P7 7 1 2 p 97->97 977->977 53->39 535->391
solved P7 7 1 2 N 49->45 494->453 40->65 401->655
solved P7 7 1 2 delta 11->10 115->107 33->37 330->373
solved P8 8 2 3 p 11->11 112->112 34->89 342->891
solved P8 8 2 3 N 48->45 481->458 51->60 517->604
solved P8 8 2 3 delta 54->41 547->410 40->16 408->168
solved P9 9 0 4 p 25->25 256->256 96->65 964->653
solved P9 9 0 4 N 10->36 100->362 53->42 536->422
solved P9 9 0 4 delta 38->89 382->896 35->41 350->416
solved P10 10 1 0 p 37->37 373->373 10->52 102->522
solved P10 10 1 0 N 31->34 310->341 26->48 268->481
solved P10 10 1 0 delta 40->42 409->428 14->13 140->139
solved P11 11 2 1 p 27->27 278->278 10->56 100->567
solved P11 11 2 1 N 79->90 797->906 66->61 667->612
solved P11 11 2 1 delta 40->10 400->101 19->38 198->382
solved P12 12 0 2 p 11->11 112->112 40->62 400->626
solved P12 12 0 2 N 97->13 973->132 39->30 397->303
solved P12 12 0 2 delta 14->84 144->842 77->38 777->382
solved P13 13 1 3 p 10->10 105->105 75->36 756->363
solved P13 13 1 3 N 10->92 103->922 59->70 595->703
solved P13 13 1 3 delta 38->26 381->268 28->29 280->296
solved P14 14 2 4 p 26->26 263->263 36->10 360->106
solved P14 14 2 4 N 10->82 105->822 50->41 504->412
solved P14 14 2 4 delta 40->16 402->160 33->51 339->515
solved P15 15 0 0 p 89->89 895->895 44->14 449->143
solved P15 15 0 0 N 66->90 662->904 72->75 724->754
solved P15 15 0 0 delta 18->25 181->258 11->29 113->298
solved P16 16 1 1 p 12->12 127->127 60->17 607->177
solved P16 16 1 1 N 14->11 144->115 13->94 131->946
solved P16 16 1 1 delta 25->29 258->290 20->15 201->153
solved P17 17 2 2 p 93->93 930->930 22->57 220->576
solved P17 17 2 2 N 33->95 335->951 98->83 986->831
solved P17 17 2 2 delta 26->38 261->383 97->96 975->964
solved P18 18 0 3 p 87->87 874->874 36->65 361->650
solved P18 18 0 3 N 85->79 852->791 41->91 412->914

```

solved P18 18 0 3 delta 16->27 169->275 81->15 816->159
 solved P19 19 1 4 p 93->93 930->930 64->68 641->682
 solved P19 19 1 4 N 47->10 474->105 31->12 311->129
 solved P19 19 1 4 delta 18->16 187->163 40->38 401->389
 solved P20 20 2 0 p 10->10 100->100 42->11 420->115
 solved P20 20 2 0 N 52->11 523->113 48->81 484->815
 solved P20 20 2 0 delta 23->16 230->160 11->24 114->240
 solved P21 21 0 1 p 29->29 298->298 12->10 125->104
 solved P21 21 0 1 N 98->68 985->688 96->79 961->798
 solved P21 21 0 1 delta 28->34 285->349 27->39 271->395
 solved P22 22 1 2 p 96->96 968->968 52->31 522->313
 solved P22 22 1 2 N 95->13 958->131 10->13 106->136
 solved P22 22 1 2 delta 14->11 144->112 36->33 365->330
 solved P23 23 2 3 p 48->48 485->485 41->83 412->830
 solved P23 23 2 3 N 11->48 118->481 55->46 553->464
 solved P23 23 2 3 delta 41->70 411->706 18->22 186->221
 solved P24 24 0 4 p 65->65 658->658 83->50 836->506
 solved P24 24 0 4 N 23->47 235->476 93->28 930->282
 solved P24 24 0 4 delta 27->29 277->299 36->10 369->102
 solved P25 25 1 0 p 81->81 817->817 35->98 350->989
 solved P25 25 1 0 N 80->23 800->238 91->20 911->203
 solved P25 25 1 0 delta 24->22 244->229 20->16 209->164
 solved P26 26 2 1 p 65->65 651->651 10->86 102->867
 solved P26 26 2 1 N 81->47 819->479 24->72 244->725
 solved P26 26 2 1 delta 20->94 200->947 11->70 112->703
 solved P27 27 0 2 p 44->44 446->446 34->24 342->240
 solved P27 27 0 2 N 35->63 355->637 20->21 202->216
 solved P27 27 0 2 delta 33->74 337->744 10->14 109->143
 solved P28 28 1 3 p 66->66 661->661 22->11 228->111
 solved P28 28 1 3 N 65->18 658->182 10->32 108->323
 solved P28 28 1 3 delta 13->25 138->259 13->22 130->226
 solved P29 29 2 4 p 57->57 577->577 12->20 129->206
 solved P29 29 2 4 N 36->81 367->816 85->73 854->733
 solved P29 29 2 4 delta 17->10 173->107 28->32 288->328
 solved P30 30 0 0 p 56->56 565->565 26->10 266->109
 solved P30 30 0 0 N 11->10 114->104 92->53 929->530
 solved P30 30 0 0 delta 26->28 264->281 27->54 274->541
 solved P31 31 1 1 p 68->68 683->683 95->81 959->811
 solved P31 31 1 1 N 25->58 254->589 70->75 702->755
 solved P31 31 1 1 delta 75->41 755->418 13->93 135->930
 solved P32 32 2 2 p 27->27 272->272 10->10 100->104
 solved P32 32 2 2 N 46->62 463->623 10->22 108->221
 solved P32 32 2 2 delta 30->23 307->231 25->32 257->326
 solved P33 33 0 3 p 14->14 143->143 88->69 888->692
 solved P33 33 0 3 N 66->44 660->443 12->42 126->423
 solved P33 33 0 3 delta 33->72 335->726 69->14 693->144
 solved P34 34 1 4 p 30->30 301->301 45->25 451->255
 solved P34 34 1 4 N 14->36 146->365 23->14 236->143
 solved P34 34 1 4 delta 67->42 673->426 11->90 110->908
 solved P35 35 2 0 p 37->37 375->375 49->70 494->706
 solved P35 35 2 0 N 67->31 678->318 72->86 720->865
 solved P35 35 2 0 delta 98->17 984->170 14->86 148->866
 solved P36 36 0 1 p 72->72 723->723 57->34 578->343
 solved P36 36 0 1 N 43->70 439->706 40->97 402->971
 solved P36 36 0 1 delta 25->36 259->364 40->30 401->308
 solved P37 37 1 2 p 10->10 106->106 85->26 850->267
 solved P37 37 1 2 N 73->43 733->430 33->26 333->262
 solved P37 37 1 2 delta 10->15 109->154 19->34 192->349
 solved P38 38 2 3 p 81->81 811->811 44->69 448->699
 solved P38 38 2 3 N 78->48 781->483 83->10 831->107
 solved P38 38 2 3 delta 15->39 158->395 93->80 937->801
 solved P39 39 0 4 p 43->43 430->430 10->81 103->815
 solved P39 39 0 4 N 10->11 108->114 28->77 282->770
 solved P39 39 0 4 delta 59->24 597->240 88->52 886->524
 solved P40 40 1 0 p 63->63 630->630 10->56 102->564
 solved P40 40 1 0 N 56->70 567->709 42->45 422->459
 solved P40 40 1 0 delta 31->26 312->266 76->12 761->125
 solved P41 41 2 1 p 96->96 969->969 45->15 458->151
 solved P41 41 2 1 N 65->59 656->596 69->11 699->114
 solved P41 41 2 1 delta 22->89 221->895 27->15 271->153
 solved P42 42 0 2 p 23->23 232->232 44->11 448->110
 solved P42 42 0 2 N 16->94 164->945 77->17 774->178
 solved P42 42 0 2 delta 42->10 425->105 92->39 926->399
 solved P43 43 1 3 p 78->78 780->780 60->11 609->112
 solved P43 43 1 3 N 48->10 481->102 67->13 679->131
 solved P43 43 1 3 delta 92->35 920->352 75->20 752->204
 solved P44 44 2 4 p 94->94 941->941 10->62 105->623
 solved P44 44 2 4 N 90->77 908->777 26->24 260->244
 solved P44 44 2 4 delta 52->27 527->277 48->86 487->860
 solved P45 45 0 0 p 78->78 781->781 10->18 107->182
 solved P45 45 0 0 N 97->28 970->280 32->60 324->609

solved P45 45 0 0 delta 20->32 209->327 12->28 123->281
 solved P46 46 1 1 p 45->45 450->450 18->44 182->441
 solved P46 46 1 1 N 57->13 570->136 80->51 805->515
 solved P46 46 1 1 delta 22->17 226->174 30->38 300->384
 solved P47 47 2 2 p 89->89 892->892 86->11 866->110
 solved P47 47 2 2 N 16->90 161->907 49->23 497->236
 solved P47 47 2 2 delta 42->28 424->285 69->48 694->482
 solved P48 48 0 3 p 10->10 108->108 89->52 891->524
 solved P48 48 0 3 N 33->99 339->999 11->17 114->175
 solved P48 48 0 3 delta 19->44 194->443 24->14 243->146
 solved P49 49 1 4 p 20->20 201->201 31->37 319->370
 solved P49 49 1 4 N 85->45 850->459 64->10 640->101
 solved P49 49 1 4 delta 96->29 962->296 38->40 383->400
 solved P50 50 2 0 p 72->72 721->721 22->66 223->668
 solved P50 50 2 0 N 24->98 241->989 10->97 106->970
 solved P50 50 2 0 delta 41->31 414->318 33->30 332->304
 solved P51 51 0 1 p 15->15 155->155 80->40 809->409
 solved P51 51 0 1 N 79->71 792->715 37->81 370->814
 solved P51 51 0 1 delta 40->98 404->985 64->28 649->289
 solved P52 52 1 2 p 53->34 533->342 70->45 706->450
 solved P52 52 1 2 N 11->21 114->212 84->10 847->109
 solved P52 52 1 2 delta 14->42 146->427 28->21 282->217
 solved P53 53 2 3 p 37->37 378->378 19->88 198->887
 solved P53 53 2 3 N 40->40 409->408 10->10 108->105
 solved P53 53 2 3 delta 21->77 216->772 41->29 411->296
 solved P54 54 0 4 p 51->51 510->510 81->75 810->752
 solved P54 54 0 4 N 18->68 180->687 25->64 256->640
 solved P54 54 0 4 delta 26->84 266->844 45->23 455->239
 solved P55 55 1 0 p 34->34 347->347 96->10 962->104
 solved P55 55 1 0 N 74->77 741->771 28->34 285->341
 solved P55 55 1 0 delta 94->43 943->435 27->24 273->245
 solved P56 56 2 1 p 46->46 465->465 54->76 549->769
 solved P56 56 2 1 N 10->36 100->360 19->92 192->928
 solved P56 56 2 1 delta 15->11 156->112 26->32 266->329
 solved P57 57 0 2 p 60->60 601->601 60->93 608->934
 solved P57 57 0 2 N 10->37 100->370 38->90 383->903
 solved P57 57 0 2 delta 91->22 917->227 33->90 335->908
 solved P58 58 1 3 p 55->55 550->550 94->65 944->659
 solved P58 58 1 3 N 59->98 590->980 67->47 677->478
 solved P58 58 1 3 delta 28->16 281->167 23->26 234->260
 solved P59 59 2 4 p 48->48 483->483 36->45 361->459
 solved P59 59 2 4 N 10->35 107->352 22->52 227->520
 solved P59 59 2 4 delta 36->34 360->341 37->25 373->257
 solved P60 60 0 0 p 47->47 471->471 84->10 849->104
 solved P60 60 0 0 N 14->30 142->305 84->21 848->218
 solved P60 60 0 0 delta 18->41 185->419 42->42 421->429
 solved P61 61 1 1 p 92->92 926->926 87->70 879->707
 solved P61 61 1 1 N 18->57 183->576 14->72 140->720
 solved P61 61 1 1 delta 20->23 201->236 13->15 133->156
 solved P62 62 2 2 p 67->67 674->674 85->64 854->643
 solved P62 62 2 2 N 33->55 332->552 80->11 805->111
 solved P62 62 2 2 delta 33->34 330->342 42->10 421->102
 solved P63 63 0 3 p 10->10 101->101 19->96 194->968
 solved P63 63 0 3 N 29->61 294->610 23->89 232->897
 solved P63 63 0 3 delta 21->31 218->319 34->10 342->106
 solved P64 64 1 4 p 60->60 605->605 80->74 809->748
 solved P64 64 1 4 N 66->84 661->845 28->64 287->643
 solved P64 64 1 4 delta 35->40 355->409 28->36 287->368
 solved P65 65 2 0 p 13->13 139->139 18->10 187->101
 solved P65 65 2 0 N 62->80 621->800 10->69 107->699
 solved P65 65 2 0 delta 20->26 201->268 37->17 377->170
 solved P66 66 0 1 p 53->53 532->532 10->63 100->632
 solved P66 66 0 1 N 11->44 115->447 10->95 106->952
 solved P66 66 0 1 delta 39->28 398->289 34->11 345->112
 solved P67 67 1 2 p 49->49 499->499 69->73 693->733
 solved P67 67 1 2 N 11->22 114->229 20->11 202->111
 solved P67 67 1 2 delta 25->20 252->201 86->33 863->339
 solved P68 68 2 3 p 98->98 985->985 37->15 371->157
 solved P68 68 2 3 N 11->95 113->954 10->66 109->664
 solved P68 68 2 3 delta 28->40 289->406 15->10 157->109
 solved P69 69 0 4 p 10->10 109->109 76->35 769->359
 solved P69 69 0 4 N 36->53 366->534 53->53 532->536
 solved P69 69 0 4 delta 35->33 350->334 58->21 580->217
 solved P70 70 1 0 p 41->41 411->411 87->47 877->475
 solved P70 70 1 0 N 73->89 731->890 80->50 809->507
 solved P70 70 1 0 delta 39->36 395->363 31->33 318->332
 solved P75 75 0 0 p 73->73 730->730 10->72 100->720
 solved P75 75 0 0 N 34->37 343->378 90->24 901->242
 solved P75 75 0 0 delta 40->10 407->107 14->13 145->139
 solved P80 80 2 0 p 16->16 169->169 88->79 888->791
 solved P80 80 2 0 N 81->60 817->609 82->42 821->423

solved P80 80 2 0 delta 32->37 323->377 18->21 189->218
 solved P85 85 1 0 p 37->37 371->371 93->29 934->296
 solved P85 85 1 0 N 48->10 487->104 20->10 204->104
 solved P85 85 1 0 delta 11->20 114->207 17->33 172->331
 solved P90 90 0 0 p 10->10 103->103 57->86 576->863
 solved P90 90 0 0 N 69->60 691->601 72->65 723->656
 solved P90 90 0 0 delta 42->25 420->251 42->21 421->212
 solved P95 95 2 0 p 11->11 118->118 47->10 475->108
 solved P95 95 2 0 N 72->10 723->108 10->74 101->742
 solved P95 95 2 0 delta 19->87 196->874 28->29 283->299
 solved P100 100 1 0 p 45->45 453->453 10->73 107->739
 solved P100 100 1 0 N 61->25 619->250 25->19 259->195
 solved P100 100 1 0 delta 94->33 940->336 17->23 173->235
 solved P105 105 0 0 p 75->75 757->757 10->51 105->514
 solved P105 105 0 0 N 98->40 982->402 93->48 937->482
 solved P105 105 0 0 delta 20->45 203->450 27->39 277->392
 solved P110 110 2 0 p 95->95 959->959 16->27 166->270
 solved P110 110 2 0 N 21->57 210->575 11->52 116->524
 solved P110 110 2 0 delta 13->16 134->164 31->17 317->177
 solved P115 115 1 0 p 62->62 625->625 88->61 889->614
 solved P115 115 1 0 N 10->45 106->456 65->32 657->322
 solved P115 115 1 0 delta 11->90 111->906 28->40 284->403
 solved P120 120 0 0 p 92->92 921->921 38->17 380->179
 solved P120 120 0 0 N 56->77 564->776 40->24 408->246
 solved P120 120 0 0 delta 47->22 475->225 30->30 302->302
 solved P125 125 2 0 p 93->93 934->934 64->11 645->113
 solved P125 125 2 0 N 59->48 599->486 17->29 173->298
 solved P125 125 2 0 delta 33->36 331->366 92->28 929->281
 solved P130 130 1 0 p 78->78 782->782 19->87 198->877
 solved P130 130 1 0 N 11->46 110->460 98->75 980->753
 solved P130 130 1 0 delta 55->12 552->120 14->16 145->166
 control R1 0 - - p 40->40 400->400 11->85 112->859
 control R1 0 - - N 61->33 612->335 78->31 783->310
 control R1 0 - - delta 91->15 912->151 65->23 659->234
 control R2 0 - - p 90->90 904->904 20->62 206->625
 control R2 0 - - N 67->80 676->808 42->68 420->685
 control R2 0 - - delta 28->81 284->819 37->23 379->235
 control R3 0 - - p 77->77 779->779 33->80 338->803
 control R3 0 - - N 34->50 347->505 77->76 775->763
 control R3 0 - - delta 25->37 254->375 31->41 316->410
 control R4 0 - - p 88->88 886->886 43->58 436->581
 control R4 0 - - N 10->17 108->171 50->49 505->492
 control R4 0 - - delta 11->29 115->293 31->42 317->427
 control R5 0 - - p 61->61 617->617 28->56 285->562
 control R5 0 - - N 94->38 947->386 10->58 106->589
 control R5 0 - - delta 36->15 365->159 89->52 891->523
 control R6 0 - - p 87->87 879->879 93->69 937->696
 control R6 0 - - N 99->38 993->388 77->29 773->298
 control R6 0 - - delta 28->31 287->316 28->33 280->335
 control R7 0 - - p 36->36 365->365 64->37 642->375
 control R7 0 - - N 87->63 873->633 49->91 499->912
 control R7 0 - - delta 36->41 369->412 29->37 298->371
 control R8 0 - - p 53->53 532->532 99->11 993->112
 control R8 0 - - N 90->92 901->928 63->24 632->245
 control R8 0 - - delta 23->22 236->221 38->32 380->329
 control R9 0 - - p 11->11 110->110 55->74 558->746
 control R9 0 - - N 44->58 442->580 48->15 483->156
 control R9 0 - - delta 75->10 752->106 28->31 286->316
 control R10 0 - - p 10->10 103->103 78->47 782->471
 control R10 0 - - N 21->46 219->469 15->11 151->113
 control R10 0 - - delta 42->19 424->197 10->90 107->908
 control R11 0 - - p 10->10 108->108 40->40 406->403
 control R11 0 - - N 92->18 921->186 54->67 545->671
 control R11 0 - - delta 32->15 320->158 38->16 383->166
 control R12 0 - - p 19->19 195->195 56->19 566->191
 control R12 0 - - N 21->10 216->101 39->88 390->884
 control R12 0 - - delta 18->29 187->299 25->27 256->275
 control R13 0 - - p 33->33 334->334 49->67 491->677
 control R13 0 - - N 39->22 391->223 72->89 728->897
 control R13 0 - - delta 20->36 208->363 31->25 319->258
 control R14 0 - - p 61->61 612->612 53->43 537->436
 control R14 0 - - N 94->89 947->896 73->44 735->447
 control R14 0 - - delta 68->18 685->182 39->91 398->913
 control R15 0 - - p 42->42 422->422 51->62 511->621
 control R15 0 - - N 10->11 100->111 69->99 693->996
 control R15 0 - - delta 71->31 719->317 31->34 317->344
 control R16 0 - - p 80->80 800->800 72->11 725->110
 control R16 0 - - N 10->61 104->611 95->38 956->381
 control R16 0 - - delta 29->36 293->368 18->13 184->139
 control R17 0 - - p 80->80 802->802 57->16 570->168
 control R17 0 - - N 11->10 112->106 58->80 583->807

control R17 0 - - delta 15->38 150->381 24->35 249->356
control R18 0 - - p 34->34 340->340 47->36 479->362
control R18 0 - - N 43->44 432->448 29->94 299->942
control R18 0 - - delta 28->24 282->242 10->39 107->390
control R19 0 - - p 89->89 899->899 13->18 137->180
control R19 0 - - N 19->71 193->711 33->13 339->139
control R19 0 - - delta 36->14 365->141 41->13 417->133
control R20 0 - - p 38->38 380->380 57->84 571->846
control R20 0 - - N 59->86 595->861 88->76 881->766
control R20 0 - - delta 70->30 704->300 39->45 397->454
control R21 0 - - p 49->49 499->499 10->35 109->351
control R21 0 - - N 10->11 107->115 88->37 880->371
control R21 0 - - delta 29->39 294->391 20->18 209->183
control R22 0 - - p 21->21 219->219 74->21 741->217
control R22 0 - - N 52->29 524->295 18->53 180->534
control R22 0 - - delta 12->32 123->329 15->66 153->664
control R23 0 - - p 10->10 108->108 10->27 104->271
control R23 0 - - N 80->61 808->610 45->82 455->825
control R23 0 - - delta 39->16 396->168 17->41 171->418
control R24 0 - - p 10->10 106->106 21->89 217->895
control R24 0 - - N 10->66 104->669 75->59 759->599
control R24 0 - - delta 12->41 128->418 16->42 169->420
control R25 0 - - p 48->48 487->487 30->10 301->106
control R25 0 - - N 10->10 100->102 25->10 257->100
control R25 0 - - delta 34->14 344->142 67->29 678->292
control R26 0 - - p 16->16 163->163 91->94 919->941
control R26 0 - - N 11->68 112->683 37->12 377->126
control R26 0 - - delta 37->30 373->304 26->39 267->390
control R27 0 - - p 16->16 166->166 41->61 415->619
control R27 0 - - N 96->62 968->626 57->21 575->215
control R27 0 - - delta 15->35 155->353 34->38 340->389
control R28 0 - - p 79->79 797->797 72->70 728->709
control R28 0 - - N 37->10 378->107 11->93 115->939
control R28 0 - - delta 14->15 140->150 27->26 279->266
control R29 0 - - p 23->23 234->234 10->46 109->462
control R29 0 - - N 78->34 788->349 10->21 105->216
control R29 0 - - delta 15->37 157->378 65->40 657->409
control R30 0 - - p 83->83 830->830 71->92 711->925
control R30 0 - - N 43->59 435->593 52->19 527->196
control R30 0 - - delta 35->90 359->906 35->12 352->126
control R31 0 - - p 80->80 805->805 45->26 458->267
control R31 0 - - N 42->74 429->744 87->37 871->374
control R31 0 - - delta 62->20 628->204 41->61 412->614
control R32 0 - - p 36->36 365->365 85->10 859->103
control R32 0 - - N 37->53 370->539 23->12 232->122
control R32 0 - - delta 20->41 205->419 39->27 393->278